

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

UNIT AND MEASUREMENTS

LECTURE-7

PHYSICS

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Topics To Covered

- 1** Revision
- 2** Rounding off
- 3** Significant Figure
- 4** Scientific Notation
- 5** PYQs

*“You learn more
from failure than
from success”*

Recap From 1st Lec.

Recap:-

Division

• Kasse parata hu

• Science

Physics

Chemistry

Biology

PO

$$PO = n \cdot u$$

• PO

Fundamental PO
Basic PO

① ② ③ ④
⑤ ⑥ ⑦

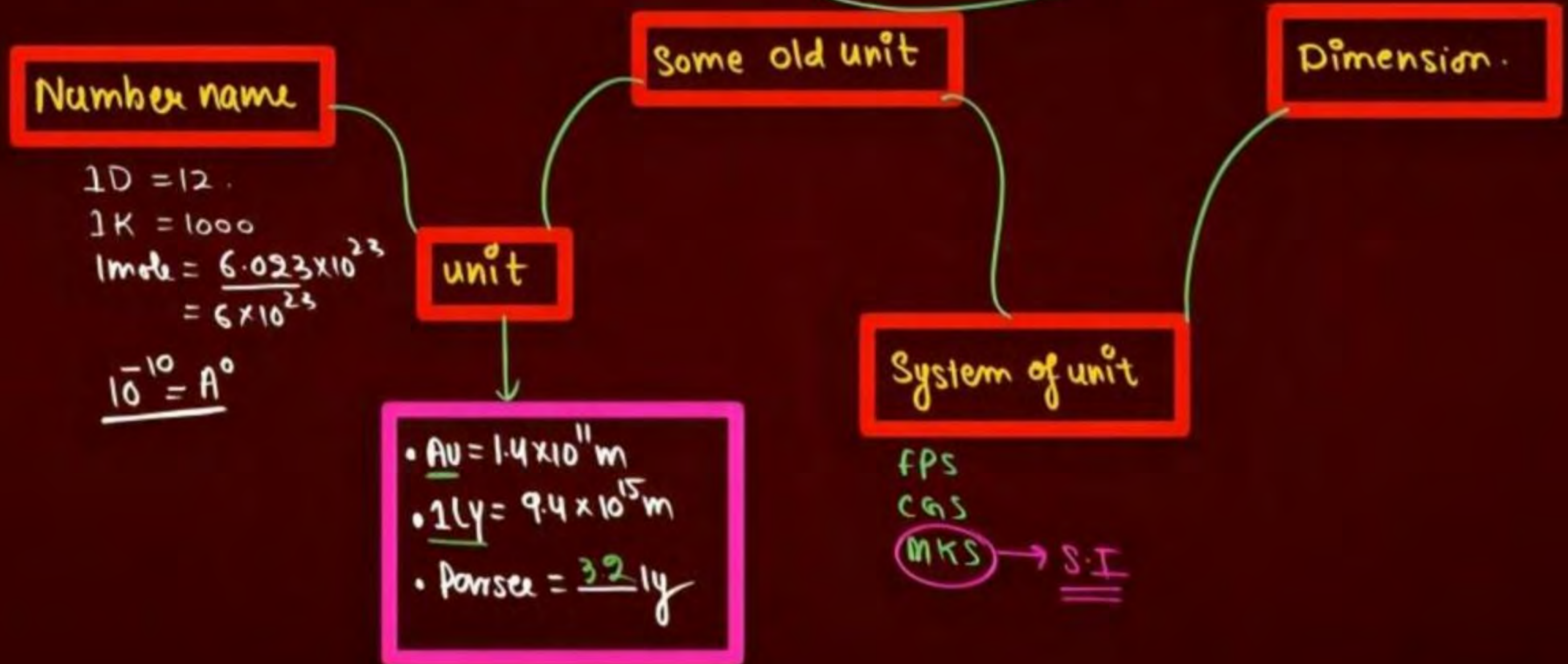
Derived PO

$$\text{Speed} = \frac{\text{distance}}{\text{time}} = \frac{m}{s}$$

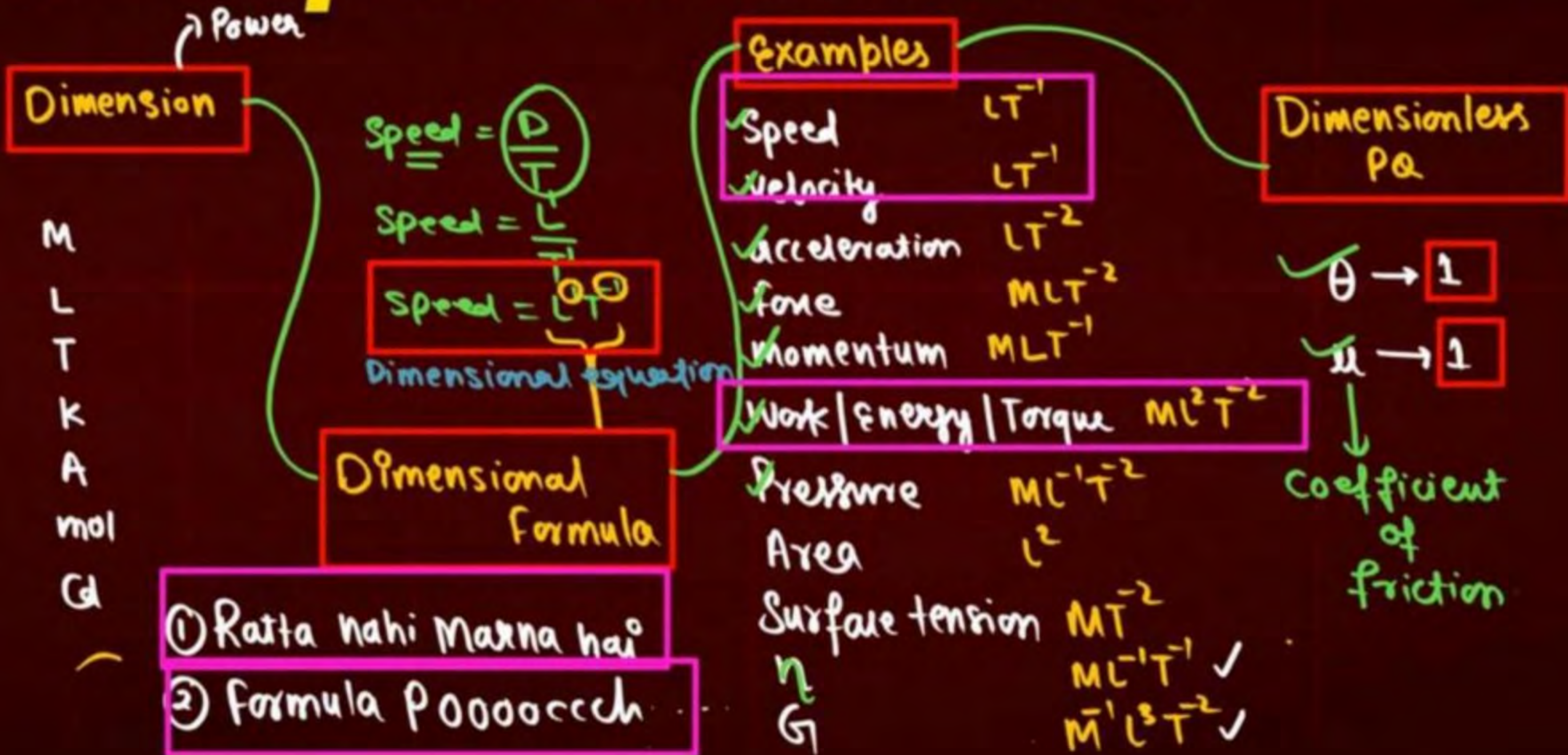
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Supplementary PO.
SA

* Number Name

Recap From 2. Lec.



Recap From 3rd Lec.



Recap From 4th Lec.

Application
of
Dimension

a ✓

b ✓

c

d

b) To check the
Correctness of
Physics Formula

Law of
Homogeneity

$$x + x = 2x$$

$$2x - x = x$$

a) To find out the
Dimensional
Formula

$$\eta \rightarrow \text{ML}^{-1}\text{T}^{-1}$$

$$G \rightarrow \text{M}^{-1}\text{L}^3\text{T}^{-2}$$

① $v = u + at$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\text{LT}^{-1} \quad \text{LT}^{-1} \quad \text{LT}^{-1}$$

② $s = ut + \frac{1}{2}at^2$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\text{L} \quad \text{LT}^{-1} \quad \text{LT}^{-1} \cdot \text{T}^2$$

③ $v^2 = u^2 + 2as$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\text{L}^2\text{T}^{-2} \quad \text{L}^2\text{T}^{-2} \quad \text{LT}^{-2} \cdot \text{L}$$

④ $F = \frac{mv^2}{r}$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\text{MLT}^{-2} \quad \text{MLT}^{-2} \quad \text{L}$$

Recap From 5th Lec.

Application of
Dimension

© To derive the
physics formula

Time period of Pendulum

• let
 $T \rightarrow m^a l^b g^c$
 $M^0 L^0 T^1 \rightarrow M^a L^b (LT^{-2})^c$
 $a=0 \quad b=\frac{1}{2} \quad c=-\frac{1}{2}$



$2T = 2\pi \sqrt{l/g}$
 $T = 2\pi \sqrt{l/g}$

Stroke law

• let
 $F \rightarrow \eta^a r^b v^c$
 $MLT^{-2} \rightarrow (ML^{-1}T^{-1})^a L^b (LT^{-1})^c$
 $a=1 \quad b=1 \quad c=1$

$F = 6\pi \eta r v$

Recap From 6th Lec.

④

conversion of
one unit system
into another
unit system

- * $1\text{m} = 100\text{cm}$
- * $1\text{kg} = 1000\text{gm}$
- * $1\text{N} = 10^5\text{dynes}$ (PVA)
- * $1\text{Joule} = 10^7\text{erg}$ (PVA)

$$n_2 = n_1 \left[\frac{M_1}{M_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$$

* Limitation of Dimension

- * we cannot derive the Dimensional const.
- * we cannot derive the formula having $(+/-)$
- * Cannot derive the formula with $\sin\theta/\cos\theta$
- * It is not unique

Rounding Off

(Topic no. 1)

To reduce the number of digit in a number

Numerical
calculation

2.4321 drop
2.432

Rule-1. → Agar too number drop hoo raha hai, 5 sa bada hai too peechu wala number 1 sa Barer Jayee ga.

2.4829 drop. $n > 5$

2.483

2.56789
2.5679

6.3487

6.349

6.5428
6.543



Rounding Off

Rule no ② Agar dropping number 5 sa kaam hai too peechu wallah number pa no effect

2.456② $n < 5$ # 2.5623①

2.456

2.5623

2.567④

2.567

3.2432④

3.2432



Rounding Off

Rule no. (3) If the dropping number = 5

→ +1 → even
 # 5.234215
 odd

5.23422

6.32835

6.3284

6.32815

6.3282

→ no change
 # 5.234225
 even

5.23422

5.2345

5.234

5.2365 → 5.236



Significant Figure

Topic no. (2)



Important topic 2025, 2024, 2023, 2022, 2021

The no. of important digits in a number. Q. Calculate the number of SF in

Rule-1. All non-zero digits are significant a given number?

3215621 SF=7

243521 SF=6

243 SF=3

24321 SF=5

Rule-2. All zero in a number is in significant

543200 SF=4

6400 SF=2

72320 SF=4

6432100 SF=5

Rule-3. the zero coming b/w non-zero digit is significant

640062 SF=6

56012 SF=5



Significant Figure

Rule-4. If number have decimal point (Right)

(P.V.O)

6.400

SF=4

6.53400

SF=6 ✓

5.23000 SF=6

6.000

SF=4

Rule-5.

(P.V.O)

If number have decimal point (left)

0.0006 SF=1

0.000562

SF=3

0.005 SF=1

0.00062

SF=2

Rule-6.

→ ~~x~~ → neglect power 10^n

$3 \times 10^{+8}$

SF=1

2×10^{-2}

SF=1

6.43×10^3
SF=3

6.023×10^{23} (SF=4)



$$\underline{642} \times 10^9 \quad \text{SF} = 3$$

$$\underline{6.421} \times 10^{-6} \quad \text{SF} = 4$$

$$3 \times 10^8 \quad \text{SF} = 1$$

Rule ⑦ If Any number

Number = infinite (Neel) - 2019

Example

<u>Number</u>	<u>S.F</u>
24321	→ 5
24000	→ 2
34051	→ 5
4.600	→ 4
0.009	→ 1
3×10^{-6}	→ 1
<u>number</u>	→ <u>infinite</u>

QUESTIONS

What is the number of significant digits in 0.005 m^2 ? 2019

$$0.005 \quad \text{SF} = 1$$

QUESTIONS

The number of significant numbers in 23.023 s: 2022

23.023

1

2

2

3

3

4

4

5 ✓

QUESTIONS

The number of significant numbers in 2.1×10^{-3} is:

2021

$$\underline{\underline{2.1}} \times 10^{-3}$$

1 ✓

2 3

3 4

4 5

QUESTIONS

The number of significant numbers in 0.0003 or 3×10^{-4} is:

1 1

2 2

3 3

4 4

$$0.000\textcircled{3}$$

 $\downarrow$
 $SF = 1$

$$\underline{3} \times \textcircled{10^{-4}}$$

 $\downarrow$
 $SF = 1$

QUESTIONS

What is the number of significant digit in 5.004 J?

5.004

SF = 4

QUESTIONS

The number 0.004 has significant figures equal to:

0.004 SF=1

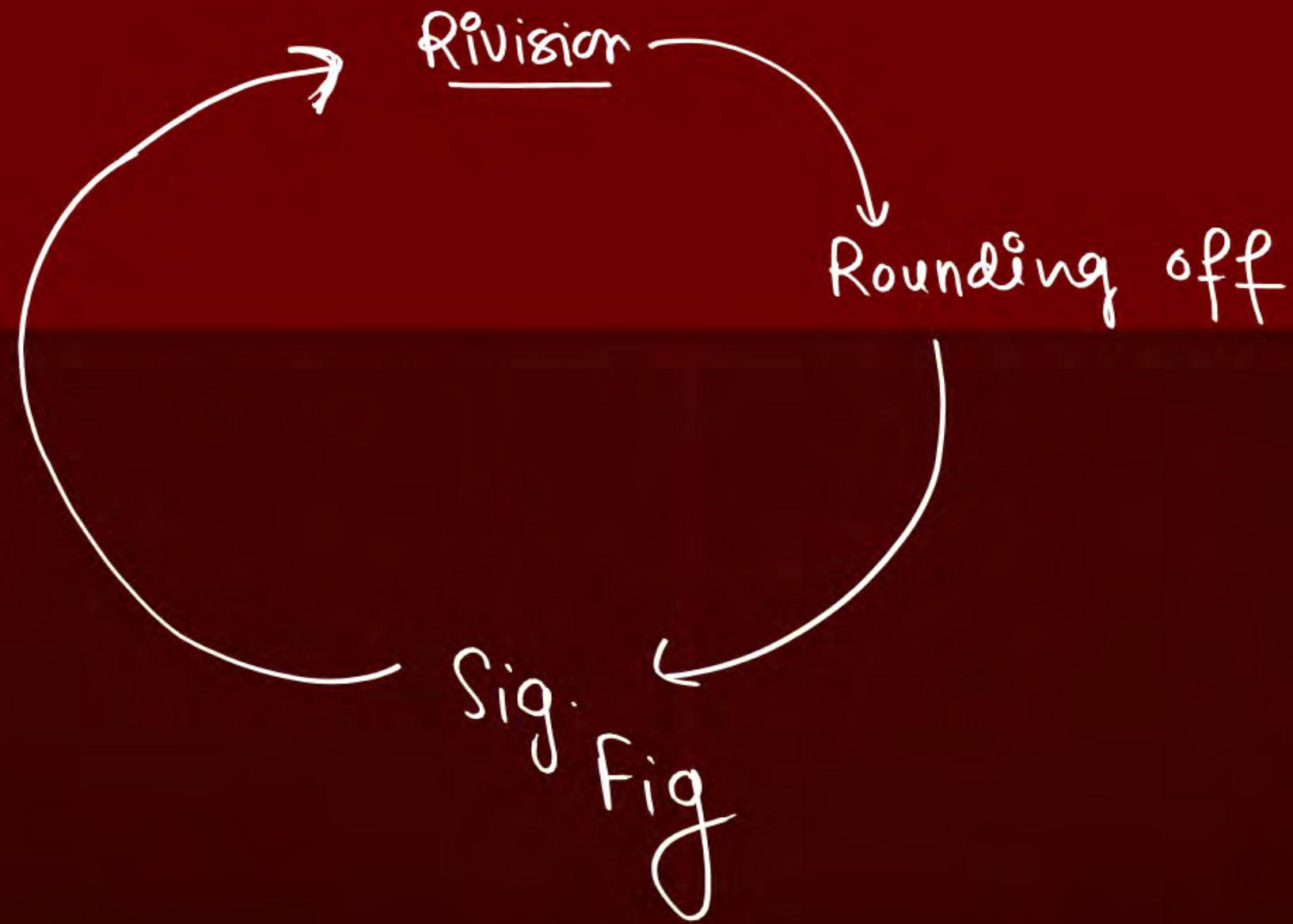
1 ✓

2

3

4

Summary



QUESTIONS

What is the dimensional formula for specific heat?

$\underline{Q} = \underline{m} S \Delta T$

$$\frac{Q}{m \Delta T} = S$$

PYQ 2021
 $Q \rightarrow$ heat energy ($M L^2 T^{-2}$)
 $m \rightarrow$ mass
 $S \rightarrow$ specific heat
 $\Delta T \rightarrow$ Temperature.

H/W

Find the dimensional formula of Strain

$$\text{Strain} = \frac{\Delta l}{l}$$

$$100\% \times 0 = \underline{0}$$

$$100\% \times 100\% \rightarrow \underline{\underline{\infty\%}}$$

Thank
You